

# Woosung Song

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## CONTACT INFORMATION

Department of Statistics  
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🌐 [Website](#)  
🐙 [Github](#)  
👤 [Google Scholar](#)

## EDUCATION

### University of California, Davis, CA

Ph.D. Candidate in Statistics, 2021 - Present (Expected graduation: June 2026)

Advisor: [Professor Hans-Georg Müller](#)

- Recipient of the Julius Blum Award (highest Ph.D. academic award in the department)
- Recipient of the Peter G. Hall Award (highest Ph.D. research award in the department)

### Seoul National University, Seoul, South Korea

B.S. in Statistics and Mathematical Sciences (Double Major), 2021

Advisor: [Professor Hee-Seok Oh](#) & [Professor Taesung Park](#)

- Grade: Summa cum laude

## RESEARCH INTEREST

Metric geometry and statistics; Manifold learning; Latent space representation; Optimal transport; Functional data analysis; Nonparametric statistics; Causal inference; Transfer learning; Brain networks

## PUBLICATIONS & PREPRINTS

1. Choi, C.<sup>†</sup>, **Song, W.**<sup>†</sup>, Müller, H.-G., & Park, B. U. (2026). Additive Fréchet regression of random objects, *Under Major Revision, The Annals of Statistics*. <sup>†</sup>Equal Contribution
2. **Song, W.**<sup>†</sup>, Zhou, H.<sup>†</sup>, Zhou, Y.<sup>†</sup> & Müller, H.-G. (2026). Non-Euclidean data analysis with metric statistics, *Harvard Data Science Review, Accepted*. [[paper](#)] [[code](#)]
3. **Song, W.** & Müller, H.-G. (2026). ADOPT: Additive optimal transport regression, *AISTATS, Accepted*. [[paper](#)][[code](#)]
4. **Song, W.** & Müller, H.-G. (2025). Inference for dispersion and curvature of random objects, *Journal of the American Statistical Association: Theory and Methods*, 1-12. [[paper](#)][[code](#)]
  - AISTATS 2026 Journal-to-Conference Track
  - Student Paper Award Finalist by ASA Nonparametric Statistics Section, JSM 2024
5. **Song, W.**, Oh, H.-S., Cheung, K. & Lim, Y. (2024). Multi-feature clustering of step data using multivariate functional principal component analysis, *Statistical Papers*, 65(4), 2109-2134. [[paper](#)][[code](#)]
6. Kim, H.<sup>†</sup>, **Song, W.**<sup>†</sup>, Choo, W., Lee, S., ..., Park, T., & Jang, J.-Y. (2023). Development, validation, and comparison of a nomogram based on radiologic findings for predicting malignancy in intraductal papillary mucinous neoplasms of the pancreas: An international multicenter study, *Journal of Hepato-Biliary-Pancreatic Sciences*, 30(1), 133-143. [[paper](#)]

7. Kang, J., Lee, C., **Song, W.**, Choo, W., ..., Park, T., & Jang, J.-Y. (2020)  
Risk prediction for malignant intraductal papillary mucinous neoplasm of the pancreas:  
logistic regression versus machine learning,  
*Scientific Reports*, 10, 20140. [[paper](#)]

## TALKS

### ADOPT: Additive optimal transport regression

- The 29th International Conference on Artificial Intelligence and Statistics (AISTATS), Tangier, Morocco. (May 2026)

### Inference for dispersion and curvature of random objects

- AISTATS, Journal-to-Conference Track, Tangier, Morocco. (May 2026)
- Joint Statistical Meetings (JSM), Topic-contributed Paper Session, Portland, OR. (Aug 2024)
- Statistics in the Age of AI, George Washington University, DC. (May 2024)
- Princeton Machine Learning Theory Summer School, Princeton University, NJ. (Jun 2023)

### Multi-feature clustering of step data using multivariate functional principal component analysis

- Fall Korean Statistical Society Conference, University of Seoul, South Korea. (Nov 2019)

## HONORS & AWARDS

|                                                                                               |      |
|-----------------------------------------------------------------------------------------------|------|
| IMS Hannan Graduate Student Travel Award, <i>Institute of Mathematical Statistics</i>         | 2026 |
| Peter G. Hall Award, <i>Department of Statistics, UC Davis</i>                                | 2025 |
| Best Presentation Award, <i>ASA Nonparametric Statistics Section</i>                          | 2024 |
| Student Paper Award Finalist, <i>ASA Nonparametric Statistics Section</i>                     | 2024 |
| Travel Award, <i>Princeton Machine Learning Theory Summer School</i>                          | 2023 |
| Julius Blum Award, <i>Department of Statistics, UC Davis</i>                                  | 2022 |
| Excellent Tutoring Award, <i>Seoul National University</i>                                    | 2020 |
| Dean's List, <i>College of Natural Science, Seoul National University</i>                     | 2020 |
| 3 <sup>rd</sup> Prize, <i>Poster presentation, Fall Korean Statistical Society Conference</i> | 2019 |

## TEACHING EXPERIENCE

### University of California, Davis Associate Instructor

- STA103: Applied Statistics for Business and Economics (Summer 2025)  
(Overall Rating: 4.75/5.0)

### Teaching Assistant

- STA010: Statistical Thinking (Fall 2021, Spring 2022)
- STA142A: Statistical Learning I (Winter 2022)
- STA141B: Data & Web Technologies for Data Analysis (Fall 2022)
- STA206: Statistical Methods for Research I (Fall 2022)
- STA100: Applied Statistics for Biological Science (Winter 2023)
- STA013: Elementary Statistics (Spring 2023)
- STA103: Applied Statistics for Business and Economics (Spring 2025)
- STA104: Nonparametric Statistics (Fall 2025)
- STA138: Categorical Data Analysis (Winter 2026)

## MENTORING

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|---------------------------------------------------------------------------------|----------------|
| Muqing Cui (now Ph.D. student at UC Davis)<br>Project: Fréchet variance process | 2023 - Present |
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|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| SERVICE                    | <b>Referee</b><br>Journal of the American Statistical Association, Biometrika, Harvard Data Science Review                                                                                                                     |                                                                      |
|                            | <b>Conference Organization</b><br><ul style="list-style-type: none"> <li>Chair, “Functional Data Analysis: Regression, Latent Structure, and Inference” session, Joint Statistical Meetings, Boston, MA. (Aug 2026)</li> </ul> |                                                                      |
| PROFESSIONAL<br>EXPERIENCE | <b>Capital One, Plano, TX</b><br>Data Science Ph.D. Intern                                                                                                                                                                     | Jun 2024 - Aug 2024                                                  |
|                            | <ul style="list-style-type: none"> <li>Developed explainable AI (XAI) models to estimate risk of account-level auto loans, ensuring transparency and optimized performance.</li> </ul>                                         |                                                                      |
|                            | <b>Republic of Korea Army, South Korea</b><br>Sergeant                                                                                                                                                                         | Feb 2016 - Nov 2017                                                  |
| SOFTWARE                   | <b>frechet</b><br>Statistical Analysis for Random Objects and Non-Euclidean Data (R package on <a href="#">GitHub</a> ).                                                                                                       |                                                                      |
| REFERENCES                 | <b>Dr. Hans-Georg Müller</b><br>Distinguished Professor, Department of Statistics, University of California, Davis                                                                                                             | ✉ <a href="mailto:hgmueeller@ucdavis.edu">hgmueeller@ucdavis.edu</a> |
|                            | <b>Dr. Jane-Ling Wang</b><br>Distinguished Professor, Department of Statistics, University of California, Davis                                                                                                                | ✉ <a href="mailto:janelwang@ucdavis.edu">janelwang@ucdavis.edu</a>   |
|                            | <b>Dr. Mina Karzand</b><br>Assistant Professor, Department of Statistics, University of California, Davis                                                                                                                      | ✉ <a href="mailto:mkarzand@ucdavis.edu">mkarzand@ucdavis.edu</a>     |